

MERZIFON, Turkey

WMO: 170820

Lat: 40.836N

Lon: 35.526E

Elev: 535

StdP: 95.06

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-7.2	-12.2	1.4	-8.5	-10.1	1.7	-6.5	10.1	3.2	8.9	3.0	0.9	10	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.0	34.0	19.7	32.0	19.3	30.2	18.7	21.3	30.7	20.4	29.0	19.7	28.0	5.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.9	25.7	17.2	13.1	24.5	16.5	12.5	23.9	64.1	30.5	61.0	29.1	58.3	28.1	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.7	8.8	-13.2	38.3	4.4	1.9	-16.4	39.7	-18.9	40.8	-21.4	41.8	-24.6	43.2		
(5)			-13.8	23.0	4.3	1.2	-16.9	23.8	-19.5	24.6	-21.9	25.3	-25.0	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	1.5	3.4	6.8	11.3	15.9	19.4	22.1	22.6	18.8	13.8	7.4	3.2
	DBStd	8.19	4.74	4.35	4.21	3.86	3.50	2.75	2.51	2.51	3.18	3.75	3.82	3.99
	HDD10.0	919	264	188	117	31	2	0	0	0	0	9	96	212
	HDD18.3	2600	522	419	358	213	90	19	2	1	32	147	329	468
	CDD10.0	1731	1	2	17	69	185	283	376	389	264	126	17	2
	CDD18.3	369	0	0	0	1	14	52	119	132	46	6	0	0
	CDH23.3	4376	0	0	6	52	285	647	1329	1388	556	113	0	0
	CDH26.7	1638	0	0	1	6	74	207	543	581	199	27	0	0
Wind	WSAvg	2.9	1.9	2.4	3.0	2.9	2.9	3.5	4.3	4.5	3.5	2.5	1.6	1.7
Precipitation	PrecAvg	406	38	28	33	51	56	45	19	12	20	31	31	45
	PrecMax	654	81	76	68	87	129	86	65	42	64	160	92	92
	PrecMin	299	4	5	1	14	22	12	0	0	0	0	2	9
	PrecStd	73	22	16	16	19	22	22	20	12	17	30	21	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	18.2	23.5	27.2	31.2	34.0	37.1	36.9	34.5	29.8	21.2	15.9
		MCWB	8.9	9.8	13.0	15.4	17.6	19.0	20.3	19.9	18.2	16.6	12.8	10.4
	2%	DB	12.0	15.6	20.0	24.9	28.8	31.0	34.0	33.9	31.2	26.2	18.9	13.1
		MCWB	7.8	9.2	11.4	14.5	17.3	18.7	20.1	19.9	17.7	15.3	11.9	8.8
	5%	DB	10.0	13.1	17.4	22.2	26.5	29.0	31.8	31.9	28.9	23.8	16.2	11.1
		MCWB	6.6	8.0	10.0	13.4	16.4	18.4	19.5	19.8	17.5	14.7	10.6	7.7
	10%	DB	8.0	10.9	15.0	19.9	24.2	27.1	29.8	30.0	26.6	21.0	14.1	9.1
		MCWB	5.6	6.9	9.3	12.5	15.8	17.7	18.9	19.1	16.9	13.8	9.6	6.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	11.1	14.0	17.1	20.0	21.1	22.9	22.3	20.3	18.0	14.1	11.0
		MCDB	13.7	16.7	21.7	24.8	28.8	30.5	32.5	32.2	30.0	26.4	19.7	15.3
	2%	WB	8.0	9.4	11.8	15.2	18.2	19.8	21.4	21.3	19.2	16.3	12.3	9.2
		MCDB	11.6	14.6	18.3	22.5	26.1	28.2	30.7	30.8	27.8	23.3	17.4	12.6
	5%	WB	6.7	8.3	10.5	13.9	17.1	18.9	20.3	20.4	18.2	15.3	11.1	8.1
		MCDB	9.4	13.0	16.4	21.1	24.7	27.1	28.8	29.1	26.1	21.5	15.7	10.8
	10%	WB	5.5	7.0	9.3	12.8	16.2	18.1	19.5	19.7	17.4	14.5	9.8	6.9
		MCDB	7.6	10.4	14.7	19.0	23.4	25.8	27.7	27.9	24.8	20.0	13.8	8.9
Mean Daily Temperature Range	5% DB	MDBR	8.5	10.6	12.0	13.9	14.8	14.7	15.2	15.0	14.8	12.7	11.7	8.3
		MCDBR	11.5	14.7	17.2	18.7	18.8	17.8	18.9	18.0	19.1	17.4	15.0	11.2
		MCWBR	8.0	9.3	9.8	10.1	9.2	7.9	8.0	7.3	8.5	8.4	9.1	7.8
	5% WB	MCDBR	10.8	13.9	15.5	16.7	16.8	16.1	16.3	15.8	16.4	14.7	13.6	10.0
		MCWBR	7.7	9.0	9.5	9.8	8.9	7.7	7.3	7.0	7.9	7.5	8.6	7.1
Clear-Sky Solar Irradiance	taub	0.320	0.351	0.394	0.435	0.410	0.386	0.393	0.397	0.372	0.369	0.332	0.319	
	taud	2.351	2.284	2.192	2.108	2.257	2.360	2.352	2.325	2.393	2.365	2.428	2.406	
	Ebn at Noon	827	850	849	837	868	889	878	864	865	820	810	799	
	Edh at Noon	86	107	131	153	135	122	122	106	106	98	80	76	
All-Sky Solar Radiation	RadAvg	1.75	2.65	3.65	4.91	5.94	6.78	6.97	6.23	4.82	3.19	2.20	1.52	
	RadStd	0.17	0.39	0.35	0.52	0.54	0.53	0.35	0.32	0.33	0.28	0.25	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.49	N/A	N/A	N/A	N/A	+0.51	-97	-131	N/A
(47) Regional (7 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	-82	-143	+106

Nomenclature: See separate page